```python
import pandas as pd
import numpy as np

data = {
    "Order_ID": [
        "ORD101", "ORD102", "ORD103", "ORD104",
        "ORD105", "ORD106", "ORD102", "ORD108",
        "ORD109", "ORD110", "ORD111", "ORD112"
    ],
    "Product": [
        "Laptop", "Mouse", "Phone", "Keyboard",
        "Monitor", "Tablet", "Mouse", "Printer",
        "Laptop", "Headphones", "Phone", "Monitor"
    ],
    "City": [
        "Chennai", "Bengaluru", "Hyderabad", "Mumbai",
        "Chennai", "Delhi", "Bengaluru", "Mumbai",
        "Delhi", "Hyderabad", "Chennai", "Delhi"
    ],
    "Quantity": [1, 3, 2, 2, 1, 2, 4, 1, 2, 3, 1, 2],
    "Unit_Price": [
        65000, 500, 25000, 1500, 12000, 18000,
        500, 9500, 65000, 2000, np.nan, 12000
    ]
}

df = pd.DataFrame(data)

df.loc[9, "City"] = np.nan

df["Revenue"] = df["Quantity"] * df["Unit_Price"]

print(df)
print("Number of records:", len(df))
```

```
    Order_ID     Product       City  Quantity  Unit_Price   Revenue
0    ORD101      Laptop    Chennai         1     65000.0   65000.0
1    ORD102       Mouse  Bengaluru         3       500.0    1500.0
2    ORD103       Phone  Hyderabad         2     25000.0   50000.0
3    ORD104    Keyboard     Mumbai         2      1500.0    3000.0
4    ORD105     Monitor    Chennai         1     12000.0   12000.0
5    ORD106      Tablet      Delhi         2     18000.0   36000.0
6    ORD102       Mouse  Bengaluru         4       500.0    2000.0
7    ORD108     Printer     Mumbai         1      9500.0    9500.0
8    ORD109      Laptop      Delhi         2     65000.0  130000.0
9    ORD110  Headphones        NaN         3      2000.0    6000.0
10   ORD111       Phone    Chennai         1         NaN       NaN
11   ORD112     Monitor      Delhi         2     12000.0   24000.0
Number of records: 12
```

```
df.to_csv("orders_dataset.csv", index=False)

print("Dataset saved successfully!")
```

```
Dataset saved successfully!
```

Case 1: Missing values in columns

Problem: Some records have missing values in City and Unit_Price.

Cause:

- Customers or employees did not enter all required details.
- Data may not have been collected correctly from the source system.
- Data imports or integrations may have failed to populate some fields.

Solution:

- Identify missing values using df.isnull().sum().
- Fill missing values when a reliable value can be determined.
- If the value cannot be recovered, investigate the source or remove the affected record when appropriate.

Case 2: Repeated Order IDs

Problem: ORD102 appears more than once in the dataset.

Cause:

- The same order may have been imported multiple times.
- A system retry may have inserted an order again.
- Multiple order items may legitimately belong to the same order.

Solution:

- Identify duplicates using duplicated().
- Verify whether repeated IDs represent duplicate orders or different products within one order.
- Remove a row only after confirming it is an actual duplicate.

```
print(df.isnull().sum())
```

```
Order_ID      0
Product       0
City          1
Quantity      0
Unit_Price    1
Revenue       1
dtype: int64
```

```
duplicates = df[df["Order_ID"].duplicated(keep=False)]

print(duplicates)
```

```
  Order_ID Product       City  Quantity  Unit_Price  Revenue
1   ORD102   Mouse  Bengaluru         3       500.0   1500.0
6   ORD102   Mouse  Bengaluru         4       500.0   2000.0
```

Case 3: Find the highest-priced product

Problem: The business needs to identify the product with the highest unit price.

Cause:

- Manually comparing prices across many products is time-consuming.
- A missing price can affect analysis if it is not handled properly.
- Incorrect data types or inconsistent prices can produce unreliable results.

Solution:

- Use max() to find the highest price.
- Use idxmax() to locate the corresponding row.
- Validate prices and exclude missing values from the comparison.

Case 4: Group revenue by city

Problem: The business needs to know how much revenue each city generates.

Cause:

- Sales are recorded across different locations.
- Revenue is not summarized by city in the original dataset.
- Missing prices or quantities can lead to incomplete revenue calculations.

Solution:

- Calculate revenue using quantity multiplied by unit price.
- Group the records by city using groupby().
- Sum the revenue for each city.

- Investigate missing cities, prices, and quantities before final reportin

```
max_index = df["Unit_Price"].idxmax()

print("Highest-priced product:", df.loc[max_index, "Product"])
print("Maximum price:", df["Unit_Price"].max())
```

```
Highest-priced product: Laptop
Maximum price: 65000.0
```

```
city_revenue = df.groupby("City")["Revenue"].sum()

print(city_revenue)
```

```
City
Bengaluru      3500.0
Chennai       77000.0
Delhi        190000.0
Hyderabad     50000.0
Mumbai        12500.0
Name: Revenue, dtype: float64
```